



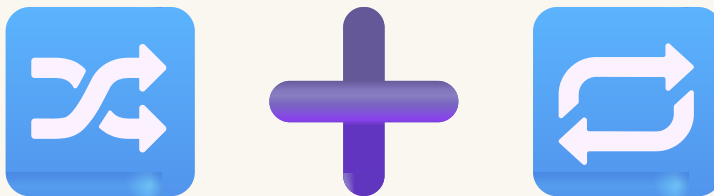
AND & OR Conditions

Smarter decisions

AND와 OR로 더 똑똑한 판단 (영어코딩)

BIG PICTURE

Two tools, used together



and = both true. **or** = at least one true.

AND는 둘 다, OR는 하나만 참이면 돼요

HOW TO COUNT

Count the axis — one block, one number



- 1 Stand on the start block — that spot is $(0, 0, 0)$.
- 2 $x \rightarrow$ walk **right**. Add **+1** for every block.
- 3 $y \rightarrow$ jump **up**. Add **+1** for every block.
- 4 $z \rightarrow$ walk **forward**. Add **+1** for every block.

Say them in order every time: $(x, y, z) = \text{across, up, forward}$.

한 칸 움직이면 +1 — 오른쪽 x , 위로 y , 앞으로 z . 순서는 늘 (x, y, z)

TYPE IT IN

Put the numbers into Minecraft



^x
across →

^y
up ↑

^z
forward ↗

```
place STONE at (x, y, z)
```

- ✓ Open **Code Builder** (press **C** in the world)
- ✓ Type the numbers in order — **(x, y, z)**
- ✓ Press **Run ▶** — your block appears at that spot

코드 빌더(C) 열고 → (x, y, z) 순서로 숫자 넣고 → Run ▶

CONCEPT

or — either one is enough

A or B is true when **at least one** side is true.

```
no wall ahead or no wall below  
# True if EITHER path is open
```

Only false when both sides are false.

하나만 참이어도 전체가 참

CONCEPT

AND vs OR side by side



and

both true
→ true



or

at least one true
→ true

AND는 엄격하게, OR는 너그럽게

CODE

OR to keep moving

```
if not agent.detect(BLOCK, FORWARD) or \
    not agent.detect(BLOCK, DOWN):
    agent.move(FORWARD, 1)
```

Move if the front **or** the way below is clear.

앞 또는 아래가 열려 있으면 전진

CODE

AND and OR in one loop

```
while not agent.detect(BLOCK, DOWN):  
    if agent.detect(BLOCK, FORWARD) and agent.detect(BLOCK, RIGHT):  
        agent.turn(LEFT_TURN)  
    elif not agent.detect(BLOCK, FORWARD) or not agent.detect(BLOCK, LEFT):  
        agent.move(FORWARD, 1)  
    else:  
        agent.turn(RIGHT_TURN)
```

Corner → turn left. Open front or left → move. Else → turn right.

WATCH IT!

OR: either branch is enough

front open

front , below open

front clear OR below clear?

✓ true → agent.move(FORWARD, 1)

✗ false → turn instead

▸ Agent does

PREDICT 🤖

Does it move?

Predict first — then click reveal

```
if not agent.detect(BLOCK, FORWARD) or \
    not agent.detect(BLOCK, DOWN):
    agent.move(FORWARD, 1)
# front BLOCKED, but below is open
```

Reveal output 🔍 reveal

VOCAB

New words today

- or** true when at least one condition is true
- and** true only when both conditions are true
- either** one side or the other is enough
- order** which check the code tries first

또는

그리고

둘 중 하나

순서



DEBUG

What's wrong with this code?

Goal: move when front **OR** below is clear.

buggy

```
if not agent.detect(BLOCK, FORWARD) and \  
    not agent.detect(BLOCK, DOWN):  
    agent.move(FORWARD, 1)
```

Think first — then click next.



DEBUG

The bug

Bug: it uses `and`, so it only moves when *both* are clear. The goal needs *either* one — swap `and` for `or`.

fixed

```
if not agent.detect(BLOCK, FORWARD) or \  
    not agent.detect(BLOCK, DOWN):  
    agent.move(FORWARD, 1)
```

YOUR TURN!

Use AND + OR

- ☒ Open the Week 4 world + Code Builder
- ☒ Use at least one **and** condition
- ☒ Use at least one **or** condition
- ☒ Watch the branch order — check the right side first
- ☒ Send your walkthrough video on KakaoTalk



QUIZ

QUESTION 1 OF 3

When is **A or B** true?

Never.

Only when both are false.

When at least one of A or B is true.

Only when both are true.



QUIZ

QUESTION 2 OF 3

How is **and** different from **or**?

or needs both; **and** needs one.

and needs both true; **or** needs only one true.

They are identical.

and only works on numbers.



QUIZ

QUESTION 3 OF 3

Why does branch **order** matter in `if` / `elif` / `else` ?

The first true branch runs, so put the more specific check first.

All branches run together.

Python runs them bottom to top.

Order never matters.

QUIZ

Your score

0 / 3



Re-read the slides. You'll get it.

↺ retake



AND + OR done!

Next: build a mini cube puzzle.

다음 주: 미니 큐브 퍼즐 만들기